



TILENCORE-GX72 CARD USER'S GUIDE

Featuring the TILE-Gx72 Processor

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EZCHIP SEMICONDUCTOR, INC.

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CHAPTER 1 OVERVIEW

Thank you for purchasing the TILEncore-Gx72™ Card featuring EZchip's revolutionary multicore Tile Processor™.

Note: In this document, "TILEncore-Gx72" is used to refer to various versions of the card.

1.1 In this Release

The following items, shown in [Figure 1-1](#), are included:

- TILEncore-Gx72, featuring the TILE-Gx72™ processor. The card comes installed with the following version of the processor:
 - TILE-Gx8072™ (Series 8000)
- Standard USB cable micro-to-A (not shown).
- RS232 cable
- Printed documentation, including the *TILEncore-Gx72 Installation Sheet* (IS105).

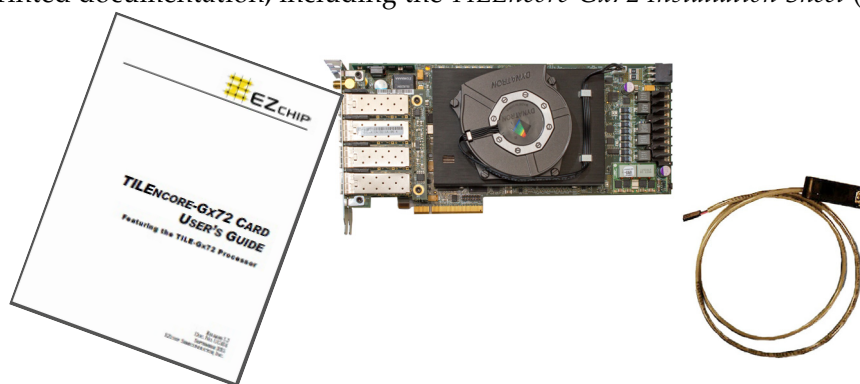


Figure 1-1: What is in the Box

Optional and Customer-Supplied Parts

- Optional cables. Customers must provide appropriate interface cables as needed. Refer to [Section 4.3 "Connecting Cables"](#) on page 17.

Interface Module Requirements

If high-speed Ethernet networking is required, customer-supplied SFP or SFP+ modules must be installed into the TILEncore-Gx72 card. Between 1 to 8 modules can be inserted, which can be a mix of SFP and SFP+ modules. Note that when the network interfaces are configured, the speed (1G/10G) must be specified to match the installed cartridge.

The modules that have been qualified for use are listed in Table 5-4, "Modules Qualified for Use," on page 25.

1.2 About this Guide

The *TILEncore-Gx72 Card User's Guide* describes the installation, setup, and features of EZchip's TILE-Gx72™ processor. This book includes the following chapters:

- [Chapter 1: Overview](#) (*this chapter*) describes the contents of this manual.
- [Chapter 2: About the TILEncore-Gx72](#) identifies the intended user for the TILEncore-Gx72 and the Card's features and supported interfaces.
- [Chapter 3: System and Card Requirements](#) specifies the power and system requirements, including the hardware and software requirements that should be followed in order to operate the Card properly.
- [Chapter 4: Installing the TILEncore-Gx72](#) provides pre-installation guidelines and proper handling instructions.
- [Chapter 5: Functional Description](#) describes some of the card's features and subsystems, specifically the card resets, LEDs, interrupts, and boot options.
- [Chapter 6: Running Programs on the TILEncore-Gx72](#) provides instructions for booting the card and running a sample program.
- [Chapter 7: Verifying that the Card Operates Properly](#) describes how to test the card to verify that it has been installed properly.
- [Appendix A: Specifications](#) provides electrical, mechanical, and environmental specifications for the Card.
- [Appendix B: Safety and Regulatory Compliance](#) lists the safety and compliance information for the TILEncore-Gx72.
- [Appendix C: Troubleshooting](#)
- [Glossary](#) provides definitions for terms used in this document.

An [Index](#) is also included.

1.2.1 Intended Audience

This book is intended primarily for hardware developers who want to use the TILEncore-Gx72 to jump start whole system designs, designers evaluating the TILE-Gx72™ processor for use in their cards, and individuals looking for a system solution. The TILEncore-Gx72 is offered separately from but in conjunction with the Multicore Development Environment™ (MDE), accompanying tools, software libraries, and software documentation. Contact your support engineer or sales representative for more information about the MDE.

1.2.2 Notation Conventions

Table 1-1 lists document notation conventions.

Table 1-1. Notation Conventions

Example	Description
Close command (File menu)	Titles in reference sections indicate the location of an item within the Graphical User Interface (GUI) for the MDE (for example, the Close command appears on the File menu).
Write In Progress (WIP) bit	Courier text indicates the names of: • A bit or bitfield, for example the CHANNEL bit • A special purpose register (SPR), for example the RSH_COORD SPR • Code sample • Application, for example the tile-monitor application • Command, for example the link command • State, for example the IDLE state
RSHIM MMIO registers	Blue and underlined text in Courier font indicates these are hypertext links to HTML files associated with the interface, or links to a specific web site.
Appendix B: Safety and Regulatory Compliance	Blue text in text font (Palatino) indicates a (cross-reference) Hypertext link to text elsewhere in the manual. If you are reading a soft-copy of this manual, you can click on the link to jump directly to the referenced section.
{this that}	Alternative required items in syntax descriptions appear within curly brackets and separated by vertical bars; read the example as this or that. One or the other is required.
[this that]	Optional items in syntax descriptions appear within brackets and separated by vertical bars; read the example as an optional this or that.
[this,...]	Optional item lists in syntax descriptions appear within brackets delimited by commas and terminated with an ellipse; read the example as an optional comma-separated list of this.
.SECTION	Commands, directives, keywords, and feature names are in text with Arial font.
<filename>	Non-keyword placeholders appear in text in italic style format and within brackets.
www.ezchip.com	Worldwide web addresses, directory names, code samples all share a courier font.
	Note: For correct operation, ... A Note provides supplementary information on a related topic. In the online version of this book, the word Note appears instead of this symbol.
	Caution: Incorrect device operation might result if ... Caution: Device damage might result if ... A Caution identifies conditions or inappropriate usage of the product that could lead to undesirable results or product damage. In the online version of this book, the word Caution appears instead of this symbol.

1.3 Technical and Customer Support

This section describes how to obtain additional information.

1.3.1 Customer Support

You can reach EZchip Semiconductor, Inc. Customer Support in either of two ways:

- Visit the EZchip® Web site at
<http://www.ezchip.com/>
- E-mail questions to
supportMC@ezchip.com

1.3.2 Product Information

You can obtain additional product information from the EZchip Web site, from the product CD-ROM, or from EZchip printed publications.

Access the EZchip Web site online at:

<http://www.ezchip.com/>

1.3.3 Related Documents

Additional documentation is available upon request. Most of the documentation listed below is included with Ezchip's MDE.

1.3.3.1 Hardware Documents

Hardware Design

- *TILE-Gx72 Processor Data Sheet (DS405)*
- *Hardware Design Guidelines for TILE-Gx Processors (WP007)*

Architecture

- *Architectural Overview for the TILE-Gx Series (UG130)*
- *TILE-Gx Instruction Set Architecture (UG401)*
- *Tile Processor and I/O Device Guide for the TILE-Gx Family of Processors (UG404)*

Cards

- *TILEncore-Gx72 Card User's Guide (UG414) (this manual)*
- Installation Sheets:
 - *Installation Sheet for: TILEncore-Gx72 with Active Heatsink (IS106)*
 - *Installation Sheet for: TILEncore-Gx72 with Passive Heatsink Installation Sheet (IS110)*
- *Board Test Kit Guide for TILE-Gx Processors (UG422)*

1.3.3.2 Software Documents

The ezchip MDE Document Index provides links to all the MDE documentation, including HTML and PDF files and additional documentation:

`${TILER_ROOT}/doc/index.html`

Note: `${TILER_ROOT}` is the root of the EZchip MDE install directory.

1.4 MDE Documentation

Note: `TILERA_ROOT` is an environment variable that points to the root of the MDE installation directory. As such it is a convenient shorthand in the documentation. It is required by many EZchip tools and examples. We recommend that you use the `tile-env` script described in *Getting Started with the Multicore Development Environment* (UG504) to set it correctly.

EZchip MDE Document Index

The EZchip MDE Document Index provides links to online copies of the MDE documentation, in both HTML and PDF formats:

```
$TILERA_ROOT/doc/index.html
```

Location of PDF Files in the Kit

The PDF versions of MDE documents can be found at this location:

```
$TILERA_ROOT/doc/pdf/
```

Man Pages and Info Pages

To view a man page for an x86 tool, run:

```
$ tile-man toolname
```

To view a man page for a Tile tool, run this on the x86 host:

```
$ tile-man toolname
```

IDE Online Help

The IDE has its own [IDE Online Documentation](#), which you can access when you use the IDE.

To get to the online help:

- Start `tile-eclipse`.

Select the menu command [Help → Help Contents](#).

- When the online help browser is displayed, select the document [IDE Online Documentation](#) in the left navigation pane, if it is not already selected.

CHAPTER 2 ABOUT THE TILENCORE-Gx72

2.1 Product Overview

The TILEncore-Gx72™ Intelligent Application Adapter card is a PCIe endpoint that enables customers to develop and test software applications on the TILE-Gx72™ processor. Physically, it is a full-height, 3/4-length, double slot PCI Express form factor with an 8-lane PCIe Gen 3 edge connector. Up to eight network interfaces are provided on the PCIe rear plate. Each of these interfaces is an SFP cage that can support either 10Gb (SFP+) or 1Gb Ethernet (SFP).

The TILEncore-Gx72 card is composed on a main board with soldered down memory, processor, two PHYs, and four SFP/SFP+ ports, and a smaller mezzanine board providing four SFP/SFP+ networking ports.

2.2 Purpose

The TILEncore-Gx72 Card serves three purposes: as an early development vehicle, as evaluation hardware for the multicore processor, and for integration into a “black box” system for developers who do not want to build their own hardware. Due to the wide variety of usage models, no software applications accompany the TILEncore-Gx72, but they can be obtained separately (see [“Customer Support” on page 4](#)).

2.3 Features

- TILE-Gx72™ processor capable of running at up to 1.2GHz.
- Four DDR3 Soldered Down memory interfaces.
- Two 4-port 10GE/GE controllers
 - Sequencing Link LED initialization
- Various interface options:
 - RS-232 (UART) port accessed via a 3-pin header (the header pins are the same as on other EZchip® PCIe cards).
 - Standard USB accessed via micro-to-A cable
- PCI Express standard, gold-fingered edge connector to host, which uses an eight-lane physical connector.
- PCIe Switch
 - On one side: two 8-lane Gen 2 connections to the TILE-Gx72.
 - On the other side: one 8-lane Gen 3 connection to host.
- User-selectable boot options, including SPI Flash ROM and host boot via PCIe.

Note: The on-board serial boot ROM is shipped with the prebooter (to configure the PCIe link), rather than with a boot image. Therefore, initial booting is expected to be via the PCIe interface.

- Temperature monitoring of the environment and the TILE-Gx72 processor.
- 12V power monitoring via software.
- Software configurable Core VDD (VID controlled).

2.3.1 Design and Layout

The TILEncore-Gx72 is comprised of two primary sections:

- The TILE-Gx processor (and its memory subsystems)
- The interface subsystem with SFP/SFP+ connectors

Figure 2-1 and Figure 2-2 depict the major card components and associations.

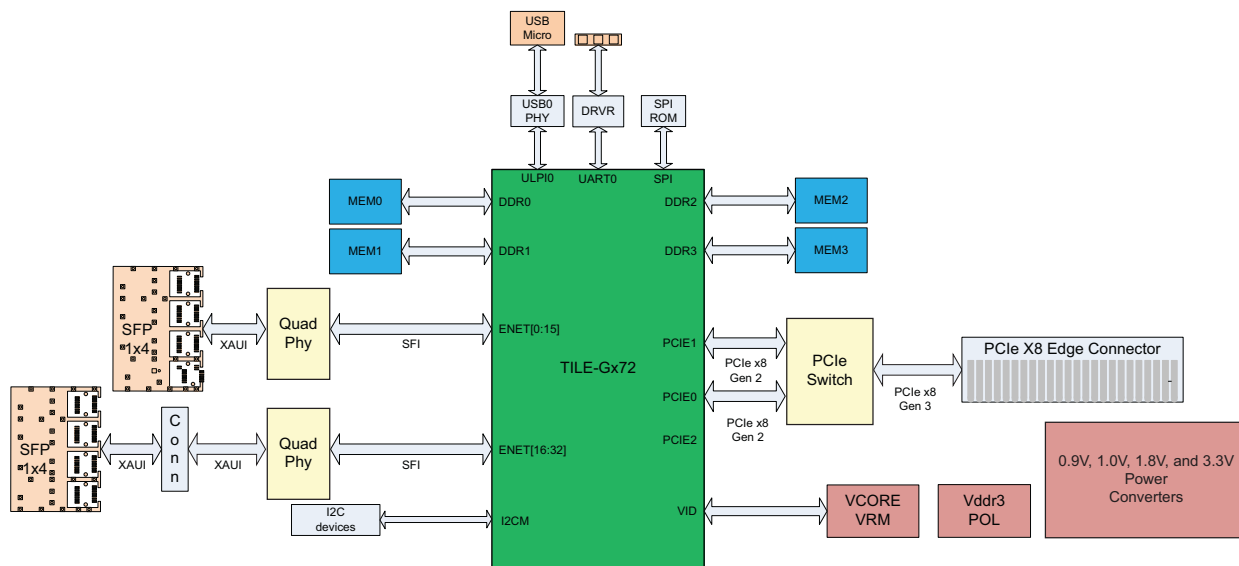


Figure 2-1: TILEncore-Gx72 Block Diagram

Figure 2-2 shows a top view of the card and identifies the major components. For locations of the LEDs, refer to Figure 7-1.

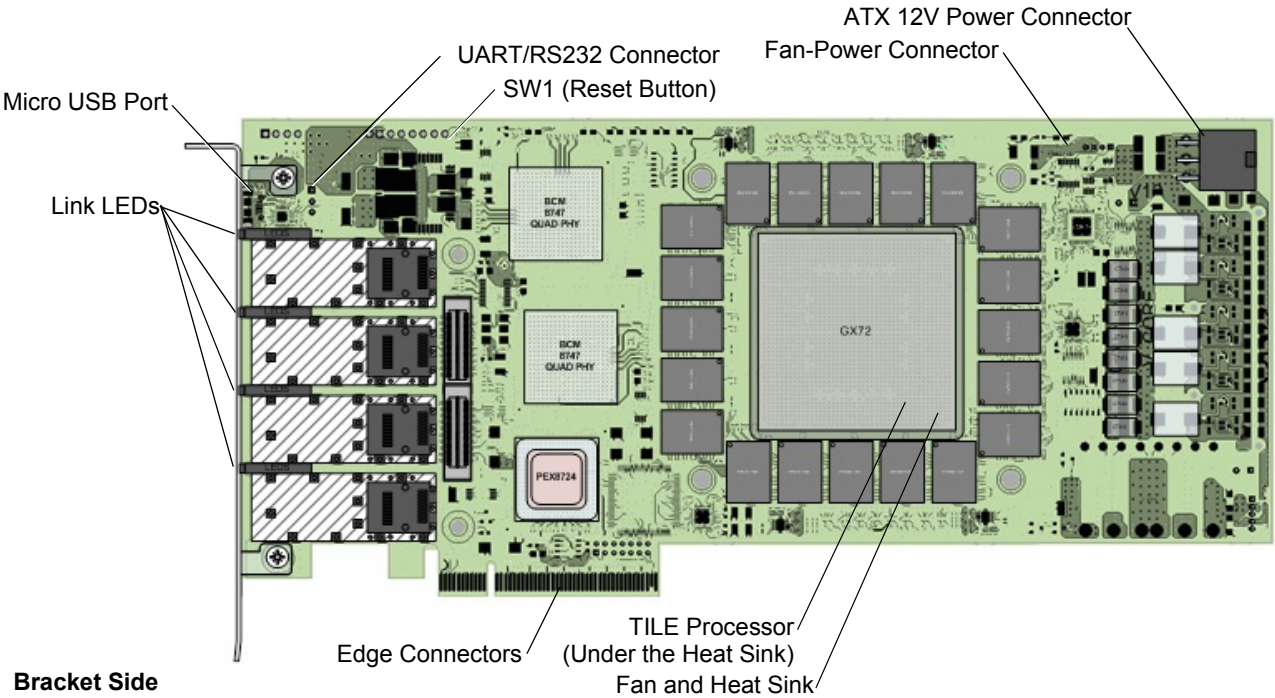


Figure 2-2: TILEncore-Gx72 — Top View

Figure 2-3 shows the bottom view of the main card and the location of the DIP switches.

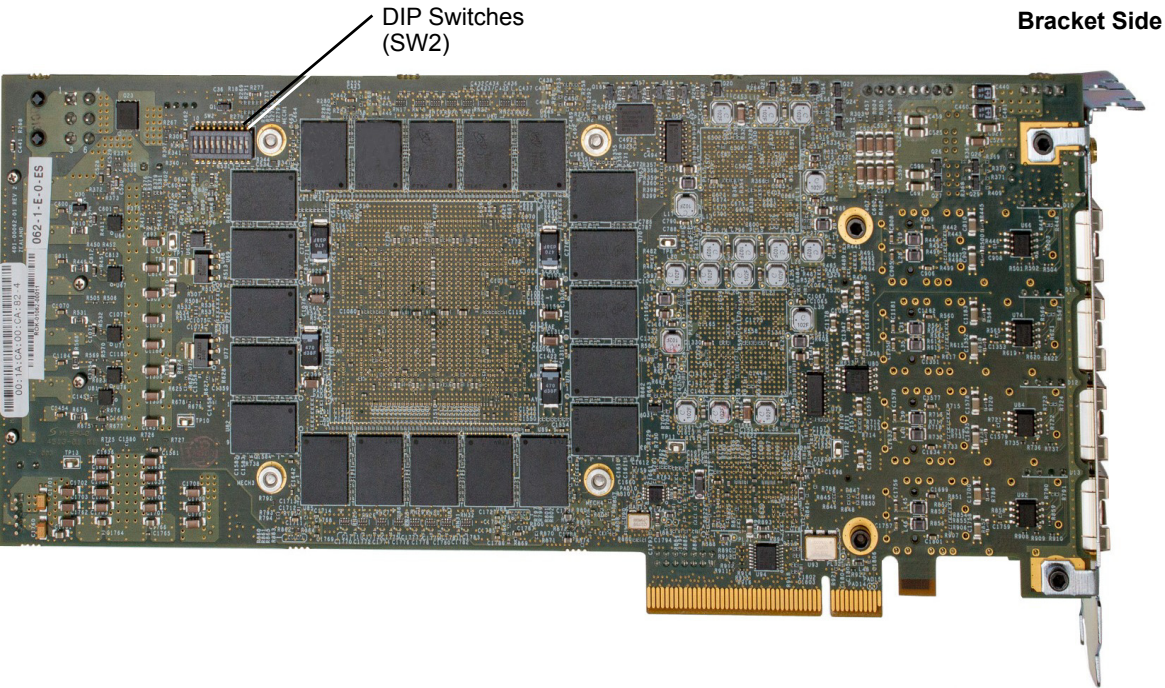


Figure 2-3: TILEncore-Gx72 — Bottom View

CHAPTER 3 SYSTEM AND CARD REQUIREMENTS

3.1 System Requirements

This section lists minimal requirements for installing and using the TILEncore-Gx72™.

3.1.1 Hardware Requirements

Minimum system hardware requirements are described in the sections that follow.

3.1.1.1 Host System

EZchip® has verified that the TILEncore-Gx72 card works in many systems. For a complete list of qualified systems, refer to “PCIe Host Requirements” in the *MDE Patch Notes* (UG918) and *MDE Release Notes* (UG908).

3.1.1.2 Mechanical Requirements

As a 3/4-length PCIe add-in card, the TILEncore-Gx72 installs in a standard x8 PCIe slot and conforms to PCIe clearance specifications for top and bottom component heights as well as specified keepouts. The system:

- Must have one PCIe x8 slot, capable of accepting a ¾-length card.
- Must have a second adjacent PCIe slot (or sufficient room equivalent to a PCIe slot) due to the width of the TILEncore-Gx72 card.
- Includes one 6-pin PC ATX power connector.

3.1.1.3 Mother Board Requirements

The TILEncore-Gx72 supports a x8 PCIe interface and requires a x8 PCIe connector. Higher bandwidth connectors are backward compatible; therefore, a x16 connector works. From an electrical standpoint, the TILE-Gx72 card trains up to Gen 3 x8.

3.1.1.4 Air Flow

For cards installed in host systems, adhere to the following recommended operating conditions:



CAUTION!

The system must provide at least 200 LFM of air flow in the passively cooled configuration:

With an active-cooled EZchip heat sink, the system airflow requirement is reduced to 0 LFM.

Note: Do not leave the PC cover open as it can disrupt air flow inside the PC. Air circulation is best with the cover closed.

3.1.2 Software Requirements

Minimum system software requirements are described in the sections that follow.

3.1.2.1 Operating System Requirements

In order for you to operate the TILEncore-Gx72, you need the following software installed on your host system:

- The TILEncore-Gx72 requires an operating system or bootrom created with MDE 4.2.2 or later. Refer to [Development Software Requirements](#) below.
- Red Hat Enterprise Linux 5 or later RHEL operating system. The MDE and drivers have been qualified on a host system running the appropriate releases of RHEL and CentOS, as specified in the *MDE Release Notes* (UG908) and/or *MDE Patch Notes* (UG918).
- The EZchip MDE includes pre-built drivers for Red Hat Enterprise Linux 6, and CentOS 6, Red Hat Enterprise Linux 5, CentOS 5. The driver source is provided; other kernels need to compile their own drivers. (EZchip provides build scripts, for example: `$TILER_ROOT/lib/modules/ tilepci-compile`). Refer to *MDE Release Notes* (UG908) and/or *MDE Patch Notes* (UG918) for a list of these pre-built drivers.

For instructions on configuring drivers, refer to “I/O Devices and Drivers” in the *Getting Started with the Multicore Development Environment* (UG504). For more information, contact supportMC@ezchip.com.

3.1.2.2 Development Software Requirements

The following software is required:

- EZchip Multicore Development Environment™ (MDE).

You must use a current copy of EZchip’s MDE— at a minimum 4.2.2 revision level. The MDE provides drivers and runtime code required to use the TILEncore-Gx72.

Refer to the *MDE Release Notes* (UG908) for specific requirements.

3.2 Power Requirements

The TILEncore-Gx72 derives its power from the power cable (ATX power cable).

- +12V from the host PC power supply. The TILEncore-Gx72 derives its power exclusively from the PCI Express 150W-ATX (6-pin) Power Connector located on the top edge of the card (see [Figure 2-2: on page 9](#)).
- It draws NO power from the PCIe “edge fingers”; all required power must be supplied from the ATX connector.

3.2.1 ATX Cable

3.2.1.1 Specifications

- +12V: Regulation tolerance = +/-5% (11.4V - 12.6V)
- Output ripple = 120 mV pp
- +5V: This is a NO-CONNECT on TILEncore-Gx72.
- The PCIe card does not tolerate “sag” or transients outside of these tolerances.

For power consumption guidelines, refer to [Section A.1 “Electrical Specifications”](#) on page 41.

Cumulative power load on the power supply must be considered when installing one or more TILEncore-Gx/TILEncore-Gx72 card(s) in a host system. Always consider the power draw required by other PCIe cards installed in the system, including the graphics card (GPU) as well.

CHAPTER 4 INSTALLING THE TILENCORE-Gx72

4.1 Pre-Installation Guidelines

This section describes steps to take when installing the TILEncore-Gx72™ card.

4.1.1 Unpacking the TILEncore-Gx72 Card

Before unpacking your TILEncore-Gx72, inspect the shipping carton for damage. If the shipping container appears damaged, notify the shipper and EZchip® immediately.

■ When you unpack your TILEncore-Gx72 card:

Carefully remove packing material. Retain the original packing in case the unit must be returned for servicing or being shipped.

4.1.2 Handling the TILEncore-Gx72 Card

4.1.2.1 Preventing Electrostatic Discharge (ESD)

In order to prevent ESD during shipment and storage, the TILEncore-Gx72 is placed in a static shielding bag, seated in an anti-static foam cushion, and placed in a box. This protection prevents vibration between the card and the box and avoids possible buildup of static electricity and ESD damage.

Before opening the box place it on an ESD-protected work area, preferably on a static dissipative table top. Workers should further protect the card by wearing a wrist strap that is connected to a ground point. The wrist strap is the primary means of controlling static charges on personnel. A non-conductive flooring is also recommended as walking across a conductive surface causes static electricity to build up.

When the card is not inserted in a PC, it should be kept in the protective bag in which it was shipped.

4.1.2.2 Preventing Damage to the TILEncore-Gx72 Card

Handling

When handling the TILEncore-Gx72 always hold it by the edges, not by any components, connectors or fixtures installed on the card. Do not flex the card as this can damage the solder connections.

4.1.3 Preparing for Installation or Removal

Tools Needed

Table 4-1 lists the tools needed for removing and installing the TIL Encore-Gx72 card in your PC.

Table 4-1. Removal and Replacement Tools

Task	Tools
Removing/installing the TIL Encore-Gx72	• Phillips screwdriver (no. 2), depending on PCI bracket style • Antistatic (ESD) wrist strap
Attaching/disconnecting the UART cable connector from the PC serial port	• Slotted (flathead) screwdriver (no. 1) • Antistatic (ESD) wrist strap

Note: It is important to use the correct size screwdriver in order to avoid damaging the screw heads.



Use caution when working inside the computer chassis as touching or bumping the card's rotating blower module can damage it.

4.2 TIL Encore-Gx72 Card Installation

This section provides installation and removal guidelines only.

Note: Some motherboards have restricted slot usage. Refer to your motherboard documentation for this information.

■ Before removing or installing any cards, follow these steps:

1. Assemble the necessary tools.
2. For a standard card installation, first shut down the host system. Ensure that all power is disconnected from the PC in which it is installed.
3. Attach an antistatic wrist guard strap to a ground point.

Note: You must NOT install or remove the TIL Encore-Gx72 card when the host system still has power, since the TIL Encore-Gx72 card is NOT hot-pluggable.

4. Read the instructions in [“Safety and Regulatory Compliance” on page 45](#) that describe how to handle the TIL Encore-Gx72 card.

■ Install a TIL Encore-Gx72 card

Follow the *TIL Encore-Gx72 Installation Sheet* (IS106) instructions provided with your TIL Encore-Gx72.

■ Remove a TIL Encore-Gx72 card:

1. Power down the host machine
2. Remove the PC's cover and place it aside.

3. Disconnect the micro-USB cable if one is installed.
4. Disconnect the UART cable and the ATX peripheral power cable.
5. Remove any SFPs prior to removing the card.
6. Loosen screws or release the lever holding the card in place.

Depending on the PC you are using, the card is held in place by either a tool-less fastening lever or a screw. You might need to use a Phillips screwdriver (no. 2) to loosen the screw securing the top of the TILEncore-Gx72 card front bracket.

7. Disengage the card from the chassis.



CAUTION! Clearance is limited between card components and the edges of the front panels of adjacent cards. Use care to avoid damaging components as you slide the card out.

8. Remove the card from the host system and place it in an antistatic bag.

4.3 Connecting Cables

EZchip provides two cables: a micro-USB cable and a UART/RS232 debug cable.

■ Connect your cables

Connect the appropriate Ethernet cables to the installed transceivers on the TILEncore-Gx72. The following cable types might be required:

Connection	Cable
SFP w/ Cu cartridge	CAT5e or CAT6 cable. For more information refer to Table 5-4 on page 25 .
SFP w/ SX cartridge	1000BASE-SX OM3 Multimode Fiber (typically orange)
SFP+ w/ SR cartridge	10GBASE-SR OM3 Multimode Fiber (typically orange)

Note: Long-range fiber connections can be used, however EZchip does not offer the SFP/SFP+ cartridges for this configuration.

4.4 Connectors and DIP Switches

This section lists TILEncore-Gx72 jumpers and DIP switch and their default settings.

4.4.1 Connectors

[Table 4-2](#) lists the TILEncore-Gx72 connectors plug settings. Default values are displayed in **bold-face** type.

Table 4-2. Connector Settings

Connector IDs	Reference Designator	Description
SFP Connector Port 0	J6	Molex Small Form Factor

Table 4-2. Connector Settings (continued)

Connector IDs	Reference Designator	Description
SFP Connector Port 1	J8	Molex Small Form Factor
SFP Connector Port 2	J11	Molex Small Form Factor
SFP Connector Port 3	J13	Molex Small Form Factor
Mezzanine Connector	J9	Samtec
USB Port	J3	USB Micro connector
150W-ATX Power Connector	J2	Six position Molex
Fan Power Connector	J1	Four position header
UART Connector	J4	Connector for UART cable. Three position, 1" header
Xilinx Programming Connector	J14	CPLD Programming Header. Fourteen position, 0.050" header

4.4.2 DIP Switch (SW2)

The TLEncore-Gx72 card has one DIP switch (SW2), as shown in [Figure 2-2](#) and listed in the following table. See also [Figure 2-3](#) on page 10 for the location of the DIP switches on the bottom of the card.

In the following table, in the “Description” section, OFF (or LOW) is equivalent to 0 and ON (or HIGH) is equivalent to 1.

SW2			
Position	Name	Defaults	Description
1	GX CONFIG_UART[0]	OFF	ON GX CONFIG_UART[0] is LOW - Protocol Mode is NOT Enabled for UART. OFF GX CONFIG_UART[0] is HIGH - Protocol Mode is Enabled for UART.
2	GX CONFIG_ROM [0]	ON	ON GX CONFIG_ROM [0] is HIGH - Boot from SPI ROM. OFF GX CONFIG_USB [0] is LOW - Do not boot from SPI ROM.
3	GX CONFIG_ROM [1]	OFF	ON GX CONFIG_ROM [1] is HIGH - Boot from I2C ROM. OFF GX CONFIG_ROM [1] is LOW - Do not boot from I2C ROM.
4	NV ROMS	OFF	ON NV ROMS Write Enabled. OFF NV ROMS Not Write Enabled.
5	PCIe Edge Finger Clock Source	OFF	ON Select Local PCIe Clock Source. OFF PCIe Edge Finger Clock Source.

SW2			
Position	Name	Defaults	Description
6	PCIe0 Endpoint	OFF	ON PCIe0 Root Complex. OFF PCIe0 ENDPOINT.
7	PCIe1 Endpoint	OFF	ON PCIe1 Root Complex. OFF PCIe1 ENDPOINT.
8		OFF	ON Software Defined, SOFTWARE_DEF_0 = HIGH. OFF Software Defined, SOFTWARE_DEF_0 = LOW.
9		OFF	ON Software Defined, SOFTWARE_DEF_1 = HIGH. OFF Software Defined, SOFTWARE_DEF_1 = LOW.
10		OFF	ON Software Defined, SOFTWARE_DEF_2 = HIGH. OFF Software Defined, SOFTWARE_DEF_2 = LOW.

4.5 Installing the TILEncore-Gx72 Drivers

■ Install drivers:

For instructions on installing drivers, refer to “Installing this Release” in the *Getting Started with the Multicore Development Environment* (UG504).

CHAPTER 5 FUNCTIONAL DESCRIPTION

This chapter describes each of the TILEncore-Gx72™ card's functional subsystems.

5.1 Power

5.1.1 Power Entry

Power is supplied from the ATX power supply connector coupled to the power supply in the chassis. Note that no power comes from the edge fingers.

The ATX connector is a standard ATX peripheral power connector that provides +12V for the TILEncore-Gx72. All other sub-voltages are generated by switching power supplies or LDO regulators on the card. This arrangement allows the PCIe card to be put into a “stand-alone” chassis where there is no connection to the edge connector.

5.1.2 ATX Power Requirements

Only +12V power, with a regulation tolerance equal to +/-5% (11.4V - 12.6V), is required from the ATX connector. The power supply must be able to supply 150 Watts, which is equivalent to ~13A through this connector.

Note: The PCIe card does not tolerate “sag” or transients outside of these tolerances.

5.1.3 Power Sequencing

On-board power sequencing is controlled by the Texas Instruments UCD9081 voltage monitor/sequencer. The UCD9081 senses when +12V has been applied and enables the switching regulators that supply power for:

- VCore (0.8V – 1.1V)
- 0.9V
- 1.0V
- 1.35V (DDR3)
- 1.8V
- 3.3V

The monitor/sequencer is configured to sequence all these voltages, listed above, up and down together. When all the voltages are between -6.5% and +25% of their nominal voltage, the power-on reset signal is de-asserted. During operations if one of the power supplies falls outside of these operating ranges, the UCD9081 shuts down the other supplies.

5.1.4 TILE-Gx Over-Temperature Protection

The National Semiconductor LM95235Q temperature sensor has an open-drain output that is pulled low when it reads the TILE-Gx die temperature above a programmed threshold. This Active LOW output travels to two locations:

- GPIO [42] of the TILE-Gx. Software uses this as an early-warning indicator that the card power is going to shut off soon, in case any process clean-up or file-logging needs to be done.
- The UCD9081 power sequencer. An overtemp event puts the sequencer into reset and all voltage rails are disabled.

5.2 Card Resets

The card supports several reset options and each reset source is able to generate a system reset, which asserts a hard reset to the TILE-Gx processor. All other devices are under software control of the Tile processor, so a hard reset of the TILE-Gx processor propagates to all the other resettable devices on the card including the XGbE PHY, USB PHY, and PCIe endpoint when the TILE-Gx is in root complex mode.

See [Table 5-1](#) for a reset propagation matrix and the subsections below for further information regarding specific device and interface reset conditions.

Note: All reset lines are active LOW.

Table 5-1. Reset Propagation

Reset Source	TILE-Gx Reset	Description
Power-On Reset	Yes	Controlled by on-card power sequencer/monitor.
Push Button Reset	Yes	Controlled by push button switch (SW1). Note that this reset button is controlled by the GX_POWER_INT_L pin, which is an input set to active low.
PCIe Reset	Yes	When the card is configured as an end point (add-in card), reset is generated by the host. This signal is an output from the card through the gold fingers when the TILE-Gx is configured to operate in Root Complex mode.

5.2.1 Power-On Reset

Power-On reset is controlled by the voltage monitor/sequencer (the Texas Instruments UCD9081), which senses when +12V has been applied and enables the switching regulators. When all the voltages are between -6.5% and +25% of their nominal voltage, the power-on reset signal is de-asserted.

The 12V supply is monitored and sequenced by the Linear Tech LTC4211 Hot Swap controller, which is enabled via +12V present on the ATX connector (via jumper J2). This Hot Swap controller fuses and ramps the +12V power supplied on the ATX connector. The “fuse” is an electronic circuit breaker that kills the input power if it reaches 12.5A. Once +12V reaches 79% of its rise, the LTC4211 enables the UCD9081 to sequence up the remaining card regulators.

The sections that follow provide information regarding specific device and interface reset conditions.

5.3 Clock Distribution

There are four clock trees on the PCIe card that are generated from a single clock driver, the IDT5V49EE904. A 25MHz oscillator provides the base input clock frequency to the IDT5V49EE904. The five output clocks are listed below. Refer to [Table 5-2](#), which contains the multipliers and divider values to produce the needed output frequencies.

- 25 MHz: This clock is distributed, single ended, to the Xilinx CPLDs
 - CPLD_REFCLK
- 125 MHz: This clock is distributed, single ended, to the GX72's Core
 - GX_CORE_REFCLK
- 156.25 MHz: This clock is distributed, single ended, to the GX72's internal Ethernet MACs
 - GX_ENET0_REFCLK
 - GX_ENET1_REFCLK
- 26 MHz: This clock is distributed, single ended, to the USB3317 USB Controller.
 - USB_ULPIO_REFCLK

The PCIe clock tree is set up to allow for the card to source the reference clock from the PCIe edge fingers, or a 100MHz PECL oscillator on-board. This multiplexing function is handled by the IDT P/N 859S0212BGILF, with the select line being sourced from the TILE-Gx72 or DIP switch, via CPLD.

The Broadcom 10G PHYs receive their reference clocks from a clock tree sourced by a 156MHz PECL oscillator. Clock distribution is done via the IDT ICS85314-11, creating four copies.

5.4 TILE-Gx Processor Connections

[Table 5-2](#) provides a list of the Tile Processor interfaces and their connections:

Table 5-2. Tile Processor I/O Connections

Description	Interfaces to...
DDR3 Memory Interface '0'	Solder-Down x16 DDR3 FBGA96 Memories (4G Bit)
DDR3 Memory Interface '1'	Solder-Down x16 DDR3 FBGA96 Memories (4G Bit)
DDR3 Memory Interface '2'	Solder-Down x16 DDR3 FBGA96 Memories (4G Bit)
DDR3 Memory Interface '3'	Solder-Down x16 DDR3 FBGA96 Memories (4G Bit)
XAUI Interface 0-3	XAUI PHY - BCM8747 "0"
XAUI Interface 4-7	XAUI PHY - BCM8747 "1"
PCIe TRIO '0' Lanes 0:7	PLX PCIe Switch - PEX8724 Port 0
PCIe TRIO '1' Lanes 0:7	PLX PCIe Switch - PEX8724 Port 1
GPIO Interface	Various I/O
UART	On-board RS-232 transceiver

Table 5-2. Tile Processor I/O Connections (continued)

Description	Interfaces to...
I2CM0	Boot/BIB ROM
I2CM1	XAUI IF PCA9555, PLX SW
I2CM2	I2C Peripheral devices (Temp sensors, power monitors, & clock control).
I2CS	N/A
SROM_SPI	On-board SPI ROM

5.5 I/O Subsystems

5.5.1 SFP+ Connectors for 1G/10G Ethernet

These connectors are located on the card and provide 10Gbps Ethernet over fiber using standard SFP+ (Small Form Pluggable for 10G) modules. This XAUI SFP+ Connector uses Ports 1-8.

Connector Detail:

- Molex SFP Connector: 20 Circuits Receptacle 30u Au Plating
- Molex P/N 74441-0010

Table 5-3. SFP Connector Summary

Pin	Signal Name	Description ^a
1	GND	Ground
2	TXFAULT	Transmitter Fault
3	TXENABLE	Transmitter Enable
4	MOD_SDA	I2C Data
5	MOD_SCL	I2C Clock
6	MOD_ABS	Module Absent
7	RS0	Rate Select 0
8	RX_LOS	Receiver Loss
9	RS1	Rate Select 1
10	GND	Ground
11	GND	Ground
12	RD-	Receiver Inverted Data Output

Table 5-3. SFP Connector Summary (continued)

Pin	Signal Name	Description ^a
13	RD+	Receiver Non-Inverted Data Output
14	GND	Ground
15	RX_PWR	Receiver Power
16	TX_PWR	Transmitter Power
17	GND	Ground
18	TD+	Transmitter Non-Inverted Data Output
19	TD-	Transmitter Inverted Data Output
20	GND	Ground

a.Per MSA SFF-8431 Specification.

Table 5-4 lists EZchip-qualified SFP/SFP+ optical modules.

Note: Use only Certified Class 1 lasers or Certified SFPs.

Table 5-4. Modules Qualified for Use

Modules	Manufacturer	Model #
SFP Optical Modules		
1Gb	Avago	AFBR-5715APZ
	Finisar	FTLF8519P2BCL
	Agilent	HFBR 5710L
SFP+ Optical Modules		
10Gb	Avago	AFBR-703SDZ
	Finisar	FTLX8571D3BCL
SFP Copper Modules		
1 Gb 1000BASE-T (non-RoHS)	Finisar	FCMJ-8521-3
1 Gb 1000BASE-T (RoHS)	Finisar	FCLF-8521-3
1.25 GBd Electrical Transceiver for 1000BASE-T, SFP, RX_LOS Disabled (RoHS-Compliant)	Avago	ABCU-5710RZ

5.5.2 UART Main Card Connector

Connector Detail:

- Reference designator J4 (UART header)
- Tyco connector

- Three circuit, .25 square pins straight post, RoHS-compliant
- Tyco P/N: 3-644456-3

Note: The UART connection is fully accessible in all configurations of the TILEncore-Gx72. Refer to [Figure 2-2: TILEncore-Gx72 — Top View on page 9](#) for an illustration of the TILEncore-Gx72.

Table 5-5. RS232 Connector Summary

Pin	Serial Cable Wire Color	Signal Name	DB-9 Female Serial RS232 Pin Number	Description
1	Green	RS232_TX	PC pin 2 (Receive Data)	TILE-GX RS232 Data Output
2	Black	GND	PC pin 5 (GND)	Power Return
3	Red	RS232_RX	PC pin 3 (Transmit Data)	TILE-GX RS232 Data Input

5.6 Miscellaneous Connections

5.6.1 Fan Power Connector

Connector Detail:

- Tyco Connector
- 4 Circuit, .25 Square Pins, Straight Post, RoHS Compliant
- Tyco P/N: 3-644456-3

Table 5-6. Fan Power Connector Summary

Pin	Signal Name	Description
1	FAN_PWM_CTRL	Fan Speed Control
2	FAN_TACH	Fan Tachometer Output
3	+12V	Fan Power
4	Ground	Power Return

5.6.2 ATX Power Connector

The ATX Power connector is used to supply +12V to the TILEncore-Gx72 card. The TILEncore-Gx72 card does not utilize any of the power pins from the PCIe Edge Connector.

Connector detail:

- Molex connector
- Header, six circuits, right angle mount, through hole, RoHS-compliant
- Molex P/N: 0457320001

Table 5-7. ATX Power Connector Summary

Pin	Signal Name	Description
1	+12V	12V Power
2	+12V	12V Power
3	+12V	12V Power
4	Sense	Sense Tied to Ground
5	Ground	Power Return
6	Ground	Power Return

5.7 TILE-Gx72 DDR3 Memory Configuration

The four memory interfaces of the TILE-Gx72 are each connected to a “soldered-down” discrete memory implementation. The devices are organized as follows:

Per TILE-Gx SHIM Density (Byte)	Device Density (Gb)	Organization	Package
4Gb(Default)	4	32 Meg x 16 x 8 banks x2	FBGA 96
8Gb	8	32 Meg x 16 x 8 banks x2 x2	FBGA 96

An additional x16 DRAM (half of which is utilized) is added to support the ECC byte.

Configuration data for these memory interfaces is stored in the Board Information Block (BIB) ROM so that the TILE-Gx72 software can configure its memory controllers to work correctly with this type of implementation.

5.8 TILE-Gx72 Processor's GPIO

Table 5-8 details the connections to the GPIO pins of the Tile Processor.

Table 5-8. TILE-Gx Processor's GPIO Assignment

Pin Name	Usage Class	Usage Detail
GPIO[0]	NC	Not Connected
GPIO[1]	NC	Not Connected
GPIO[2]	NC	Not Connected
GPIO[3]	NC	Not Connected
GPIO[4]	NC	Not Connected

Table 5-8. TILE-Gx Processor's GPIO Assignment (continued)

Pin Name	Usage Class	Usage Detail
GPIO[5]	NC	Not Connected
GPIO[6]	NC	Not Connected
GPIO[7]	NC	Not Connected
GPIO[8]	NC	Not Connected
GPIO[9]	NC	Not Connected
GPIO[10]	NC	Not Connected
GPIO[11]	NC	Not Connected
GPIO[12]	NC	Not Connected
GPIO[13]	NC	Not Connected
GPIO[14]	NC	Not Connected
GPIO[15]	NC	Not Connected
GPIO[16]	GX_XGBE_INT0	XAUI PHY Link Alarm Status Interrupt
GPIO[17]	NC	Reserved For Future Use
GPIO[18]	NC	Reserved For Future Use
GPIO[19]	NC	Reserved For Future Use
GPIO[20]	GX_XGBE_INT1	XAUI PHY Link Alarm Status Interrupt
GPIO[21]	NC	Not Connected
GPIO[22]	NC	Not Connected
GPIO[23]	NC	Not Connected
GPIO[24]	GX_XGBE_INT2	XAUI PHY Link Alarm Status Interrupt
GPIO[25]	NC	Not Connected
GPIO[26]	NC	Not Connected
GPIO[27]	NC	Not Connected
GPIO[28]	GX_XGBE_INT3	XAUI PHY Link Alarm Status Interrupt XAUI Interrupt
GPIO[29]	SOFTWARE_DEF_0	Software Defined dip switch jumper
GPIO[30]	SOFTWARE_DEF_1	Software Defined dip switch jumper

Table 5-8. TILE-Gx Processor's GPIO Assignment (continued)

Pin Name	Usage Class	Usage Detail
GPIO[31]	SOFTWARE_DEF_2	Software Defined dip switch jumper
GPIO[32]	GX_XGBE_INT4	XAUI PHY Link Alarm Status Interrupt
GPIO[33]	PLX_SW_INT_L	PLX Switch Interrupt Input
GPIO[34]	GX_PLX_TEMP_INT_L	PLX Switch over Temperature Interrupt Input
GPIO[35]	GX_QUAD_PHYB_RESET_L	BroadCom XAUI PHY 1 Reset Output
GPIO[36]	GX_XGBE_INT5	XAUI PHY Link Alarm Status Interrupt
GPIO[37]	GX_QUAD_PHYA_RESET_L	BroadCom XAUI PHY 0 Reset Output
GPIO[38]	GX_USB_ULPI0_RESET_L	Reset to CPLD for USB Controller, output, active low.
GPIO[39]	GX_PCIE_CLK_SOURCE	Select Local PCIe Clock Source or edge connectors
GPIO[40]	GX_XGBE_INT6	XAUI PHY Link Alarm Status Interrupt
GPIO[41]	USB_OC_FLAG_L	USB Over Current FLAG_L, input, active low
GPIO[42]	GX_POWER_INT_L	From Xilinx CPLD, input, active low.
GPIO[43]	GX_TEMP_INT_L	TLR Die Temperature Interrupt, input, active low
GPIO[44]	GX_XGBE_INT7	XAUI PHY Link Alarm Status Interrupt
GPIO[45]	GX_PCIE0_PERST_N	Trio Bus 0 To Xilinx CPLD, Bi-directional
GPIO[46]	GX_PCIE1_PERST_N	Trio Bus 0 To Xilinx CPLD, Bi-directional
GPIO[47]	GX_XAUI_MEZZ_PRSENT_L	Indicates XAUI 4-7 Mezzanine installed

5.8.1 TILE-Gx Processor's I²C — Master Interface

There are three master and one slave interfaces connected to the TILE-Gx Processor's I²C Master interface. The Slave interface is unused. There are 16 devices connected to the TILE-Gx72's I²C Master interface. [Table 5-9](#) lists the device address, Master Number, part type and description.

[Table 5-9](#) lists the device address, part type, and description.

Table 5-9. I²C Address and Device Map

I ² C Address 7-bit Address	Vendor and Part Number	Description
1010 100X	Atmel - AT24C512	2-wire ROM, 512 Kbits (Atmel), Boot ROM 0
0100 000X	NXP PCA9555	16 Bit GPIO Expander

Table 5-9. I²C Address and Device Map (continued)

I ² C Address 7-bit Address	Vendor and Part Number	Description
0111 000X	PLX PEX8724	PLX PCIe Switch Registers
1110 000X	Texas Instruments PCA9546A	4 Channel I2C Switch.
1100 000X	Texas Instruments UCD9081	8 Channel Power Sequencer and Monitor
1000 000X	Texas Instruments INA219	Current/Power Monitor (12V)
1101 010X	IDT IDT5V49EE904	Clock Driver Chip
0101 100X	Maxim MAX6639	TILE-Gx72 Temp 0 Sensor and Ambient & Fan Ctl
1001 100X	National Semi LM95235QQ	PLX PEX8724 Temp Sensor

5.8.1.1 I²C ROM

An I²C 2-wire serial EEPROM is available for user configuration data. It is connected to the TILE-Gx Processor's I2CM0 interface. The device is an Atmel AT24512 or equivalent 512K part.

5.8.1.2 I²C Temperature Sensor

There are three differently grouped temperature sensors on the card. One sensor for the TILE-Gx72, one sensor for the PLX PEX8724, and one to measure ambient card temperature.

The Maxim MAX6639 and National Semiconductor LM95235Q Temperature sensors together are used to monitor the temperature of the TILE-Gx72 to prevent damage to the device due to overheating. The MAX6639 is also used to control the TILE-Gx72's fan speed. The LM95235Q senses its own temperature, as well as that of the diode junction resident on the TILE-Gx72's die.

Both the MAX6639 and LM95235Q have open-drain outputs that are pulled low when it reads the TILE-Gx72 die temperature above a programmed threshold. This active low output goes to the CPLD and then on to the TILE-Gx72's GPIO.

TILE-Gx72

GPIO00 [42] of the TILE-Gx72. This should be used by software as an early warning indicator that the card power is soon to be shut off, in case there is any process clean-up or file-logging that needs to be done prior to the UCD9081 power controller beginning the power down sequence. Any housekeeping needs to be done within μ seconds.

A National Semiconductor LM95235Q Temperature sensor is used to monitor the temperature of the PLX PEX8724 to prevent damage to the device due to overheating. The LM95235Q has an open-drain outputs that will be pulled low when it reads the PLX die temperature above a programmed threshold. This active low output goes to the CPLD and then on to the TILE-Gx72's GPIO.

PLX

GPIO00 [34] of the TILE-Gx72. This should be used by software as an early warning indicator that the PLX is in thermal trouble.

5.8.2 TILE-Gx Processor SPI Interface

5.8.2.1 TILE-Gx Processor SPI FLASH Memory

In addition to the I2C 2-wire serial EEPROM, there is a SPI ROM for boot loading or other user configuration data. It is connected to the TILE-Gx72's SPI interface. The device is a Spansion S25-FL128P0XNFI01, 128Mbit part.

5.8.3 TILE-Gx External Interrupts

[Table 5-10](#) details the interrupts to the TILE-Gx.

Table 5-10. TILE-Gx Processor's Interrupt Assignment

Interrupt Name	Interrupt Source	Physical Connection
GX_POWER_INT_L	Xilinx CPLD	GX72 GPIO00[42]
GX_TEMP_INT_L	Xilinx CPLD	GX72 GPIO00[43]
GX_PLX_TEMP_INT_L	Xilinx CPLD	GX72 GPIO00[34]
GX_XGBE_INT[0:7]	BCM8747[0:1]	GX72 GPIO00[16,20,24,28,32,36,40,44]

5.8.4 PCIe Resets

PCIe fundamental reset is connected to the TILEncore-Gx72 from the PCIe edge fingers and is distributed from an on-board CPLD to the following endpoints: TILE-Gx72 TRIO0 and TRIO1, and the PLX PEX8724.

On TILEncore-Gx72, the reset functionality depends on the setup of TILE-Gx72 TRIOs, specifically whether or not they are configured as PCIe "Root Complex". When neither of the TRIO ports on the TILE-Gx72 is Root Complex, the PCIe reset signal from the edge connector is logically connected to the TILE-Gx72's hard reset signal, which causes the TILE-Gx72 to execute a Level 0 boot.

When one of the TRIO ports is configured as the Root Complex, the TILE-Gx72 has two GPIO pins (GPIO [45:46]: GX_PCIE [0:1] _PERST_N), which are connected to the Xilinx CPLD. If this signal is asserted (high), then the PCIe reset to the gold fingers can be generated by the TILE-Gx72 via either GPIO [45] or GPIO [46].

It should be noted that only ONE TILE-Gx72 TRIO can be Root Complex at a time. The TILEncore-Gx72 card does not support multiple Root Complexes through the PLX switch, so the TILE-Gx72 can only operate a Root Complex TRIO port if there is no host system connected externally with a Root Complex port.

5.9 LEDs

5.9.1 Power and Status Indicators

TILEncore-Gx72 includes seven power-related LEDs and one software-programmable red LED on the card, as listed on [Table 5-11](#). [Figure 5-1](#) shows the location of the Power On and Power OK LEDs with the card in standalone mode. Note that these LEDs do not include Port Link Status/Activity LEDs, which are controlled by the PHY (see [Figure 7-1](#)).

For more information on Link Status LEDs, refer to [Figure 2-2: TILEncore-Gx72 — Top View on page 9](#) or [Figure 7-1: SFP/SFP+ Link and Power LEDs on page 39](#).

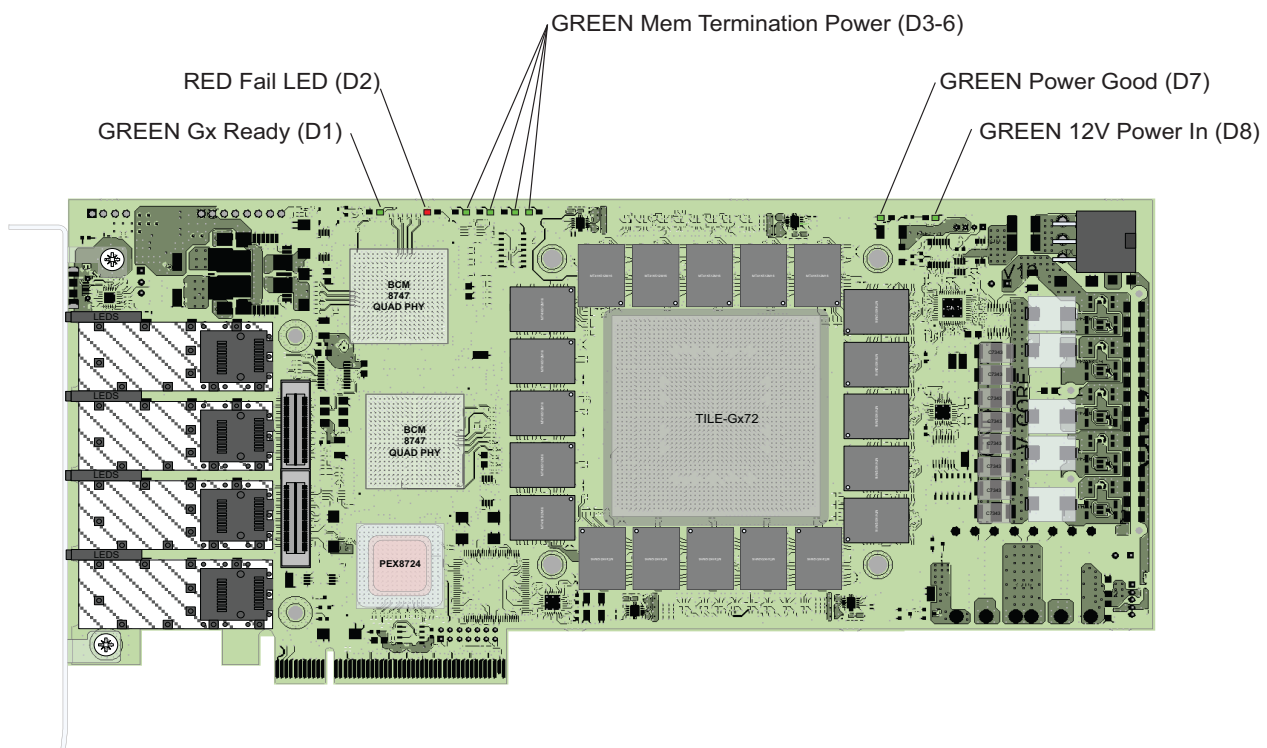


Figure 5-1: TILEncore-Gx72 Card LEDs— Top Side

Table 5-11. Card LED Definitions

Reference	Color	Label	Location	Description
D1	GREEN	GX READY	Front	This LED is tied to a CPLD Output, and will indicate that the GX is out of reset and in a “Ready” state.
D2	RED	FAIL LED	Front	This LED is tied to a CPLD Output, and is somewhat programmable in nature. Initial CPLD code will tie this to BOARD RESET so it should turn off as the software boots the TILE-Gx72.
D3-6	GREEN	Mem Term Pwr	Front	Memory Terminate Power. These LEDs
D7	GREEN	POWER GOOD	Front	This LED is tied to 3.3V to indicate that all secondary voltages are up and running on the card.
D8	GREEN	12V POWER IN	Front	The hot swap controller is enabled and +12 V is being provided to the card's on-board regulators.

CHAPTER 6 RUNNING PROGRAMS ON THE TILENCORE-Gx72

6.1 Problems with Simultaneous USB and PCIe Usage

The TILEncore-Gx72™ card supports boot and debug operations via its PCIe interface, as well as via its USB device interface. It is possible to connect both of these interfaces simultaneously, but doing so can cause undesired system behavior. This section explains why that is and how to avoid it.

The problem is related to booting the TILE-Gx™ processor, which happens whenever you run `tile-monitor` *without* the `--resume` option, or whenever you run any chip tests via the Board Test Kit (BTK). The first step of booting the TILE-Gx processor is to reset it. This operation is done automatically by the `tileusb` or `tilegxpci` drivers when you start to boot the processor via the corresponding interfaces.

When the processor is reset, all of the I/O interfaces on the chip get reset. In the case of USB, the USB device appears to be disconnected from the host; in the case of PCIe, the PCIe link between the host and the processor goes down. Soon afterward, communication is re-established.

In the case of USB, the link is brought back up automatically by the processor; in the case of PCIe, the PCIe prebooter is loaded from the SROM and brings up the link. However, the host's response to these momentary interruptions is very different. USB is designed as a hot-pluggable interface. The cabling for USB devices is generally accessible to their users, and many of those devices are connected to their host system only temporarily (such as when you download photographs from a digital camera). Because of this, the USB software stack expects devices to be disconnected and reconnected.

While PCIe devices can also be hot-pluggable, this is the exception, rather than the rule. Desktop computers, which are commonly used to host TILEncore-Gx72 cards, typically do not support hot-plugging. Thus, their PCIe software stacks do not expect PCIe devices to be disconnected.

In order to cope with the fact that rebooting (and thus resetting) a TILE-Gx processor causes the USB or PCIe connection to be interrupted, the MDE's `tileusb` and `tilegxpci` drivers perform special steps as part of a reboot operation. The driver might need to read the configuration state from the device, though state information is destroyed by the reset. Once the device has been reconnected, the configuration state is restored. The driver might also need to reconfigure the system temporarily to tolerate the upcoming disconnect event. However, these steps can only be performed if the relevant driver knows that a reset is about to happen, which brings us to the problem with simultaneous USB and PCIe connections.

Imagine what happens when a TILEncore-Gx72 card is configured into a host system and both its USB and PCIe ports are connected. Both the `tileusb` and `tilegxpci` drivers believe they are responsible for controlling the card, and both devices are willing to reboot and thus reset it. However, neither driver has any "knowledge" of the other. Thus, if you use the `tileusb` driver to reset the card, the `tilegxpci` driver does not get the opportunity to do any of the normal state-saving and disconnection preparations; the host system sees an unexpected interruption of service to one of its PCIe devices instead.

The effect of this kind of event is system-dependent. On some systems, the unexpected PCIe disconnect leads to a fatal error and a system crash. On others, the system does not crash, but the fact that the newly-reset TILEncore card has lost the configuration information programmed into it during device discovery can lead to problems with PCIe I/O, perhaps not even involving the TILEncore card. In either case, the only way to recover is to reboot the host system.

Because of this, we strongly recommend that if you have a TILEncore-Gx72 connected via both PCIe and USB, that you never reboot the card via USB.

This means that you should not run `tile-monitor` against the USB interface, and should not run any tests when using `tile-btk` against the USB interface. Rebooting such a doubly-connected card via PCIe, on the other hand, is generally safe, since the USB interface is much more tolerant of momentary interruptions.



If you have PCIe and USB connected at the same time, **you should avoid resetting the card via USB.**

If you reset the card over USB, one of two things is very likely to happen:

- If your card is in a host **that is not sensitive to the PCI link going down**, the PCI link goes down and comes back up, but the TILE-Gx72 is not programmed properly so the host cannot “talk” to the TILEncore-Gx72 via the host's PCIe bus. It is also possible that the default settings on the card might interfere with other things on the bus, which is less likely but still possible. The main symptom is that the host cannot control the TILEncore-Gx72 card via the host's PCIe bus. The only way to fix that problem is to reboot the host system.
- If your card is in a host **that is sensitive to the link going down**, the system most likely crashes and reboots. Some hosts might not be set up to reboot after a failure like that. You might need to power cycle the host in these cases.

6.2 Boot Options

The TILEncore-Gx72 card supports the following boot options:

- Boot from host via the TILE-Gx72 PCIe0 interface.
This is the standard boot option (e.g. `--dev gxpciX`).
- Boot from host via the TILE-Gx72 PCIe1 interface (e.g. `--dev gxpciX-link1`).
- Boot from USB.
- Boot from boot ROM (SPI ROM).
Requires programming of SPI ROM.
- Boot from UART.

UART port configured in `UART_PROTOCOL_MODE`.

Refer to the DIP switch configuration information in [Section 4.4 Connectors and DIP Switches](#) for settings related to boot options.

6.2.1 Booting a TILEncore-Gx72 via a PCI

```
tile-monitor --dev gxpci0 \
--hvx TLR_NETWORK=none --hvd POST=quick \
--hvx "dataplane=1-71" \
--vmlinux $TILER_ROOT/tile/boot/vmlinux \
--hvc $TILER_ROOT/tile/etc/hvc/vmlinux.hvc \
```

The `--vmlinux` argument specifies a Linux kernel to boot. Instead of booting the default kernel as shown here, you could specify a different kernel (e.g. your custom-built Linux kernel).

The `--hvc` argument specifies a hypervisor configuration file. Instead of using the default as shown here, you could specify any other hypervisor configuration file. Refer to *EZchip MDE System Programmer's Guide* for more information on how to configure an alternative hypervisor config then configure and build an alternate vmlinux kernel.

6.2.2 Booting a TILEncore-Gx72 via USB

Note: Do not to use USB to boot if the card is installed inside a host PC.

To run a program on the TILEncore-Gx72™ card for exhibition purposes, use the `--dev` option:

```
tile-monitor --dev usb0 \
--hvd POST=quick \
--hvx TLR_NETWORK=none \
--hvx "dataplane=1-71" \
```

This automatically reboots the card. If you only have one card in your system, you can leave off the `--dev usb0`, for example:

```
tile-monitor -- <executable> [<argument> ...]
```

You can also run `tile-monitor` to connect to an already-booted TILEncore-Gx72 card without rebooting it by using the `--resume` option:

```
tile-monitor --dev usb0 --resume
```


CHAPTER 7 VERIFYING THAT THE CARD OPERATES PROPERLY

7.1 Status LEDs

The TILEncore-Gx72™ card has eight Power and Health LED indicators located on top edge of the card. These are defined in [Table 5-11 on page 33](#). When a card is in good working order, powered up, and ready for operation, all seven GREEN LEDs (D1, D3-6, D7, and D8) should be lit. The RED LED (D2) should never light except very shortly after power has been applied.

Note: If power is applied, but no boot stream is available (it is waiting for the host to boot it over PCIe or USB), then this RED LED stays lit until the boot starts.

For additional information, refer to [Section 5.9 “LEDs”](#) on page 32.

7.2 Port Link Status/Activity LEDs

The LINK LEDs, located on the bracket and depicted in [Figure 7-1](#), indicate that the XAUI links are operating properly.

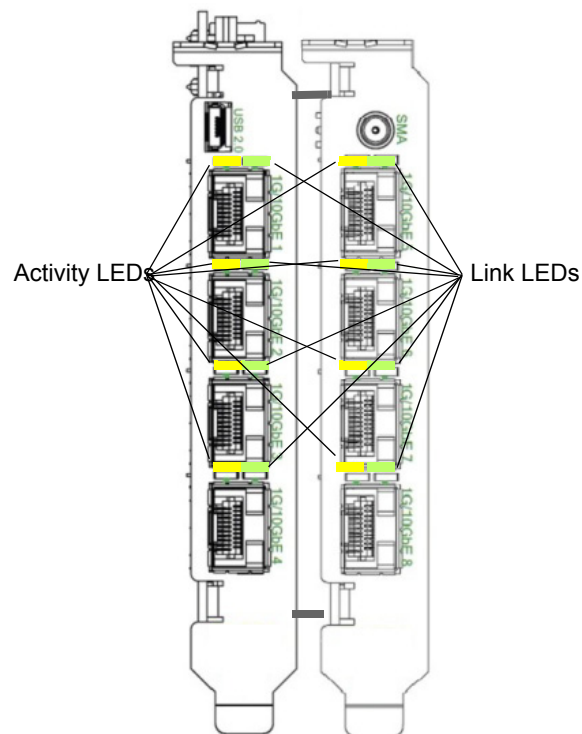


Figure 7-1: SFP/SFP+ Link and Power LEDs

■ **To test the TILEncore-Gx72:**

Run a `tile-monitor` command that executes an `echo` or an `ls` command to verify that the card is operational as follows:

```
tile-monitor --dev gxpci --ls
```

For more information on using the `tile-monitor` command, refer to *Getting Started with the Multicore Development Environment* (UG504)

APPENDIX A: SPECIFICATIONS

A.1 Electrical Specifications

Power consumption varies depending on application, but [Table A-1](#) can be used as a guide:

Table A-1. Power Consumption

Card Version	Typical	Maximum
TILEncore-Gx72 card @ 1.0 GHz	95W	117W

See also [Section 3.2 “Power Requirements”](#) on page 12.

A.2 Mechanical Specifications

The mechanical dimensions, of the card are compliant with the add-in card form factor for an 8-lane PCI Express add-in card. The TILEncore-Gx72 is a 3/4-length card that measures 9.44 inches long.

Table A-2. Physical Specifications

Card Height	11.1 cm (4.38 in)
Card Length	23.98 cm (9.44 in)
Card Width	3.7 cm (1.46 in)
Bracket Height	12 cm (4.72 in)

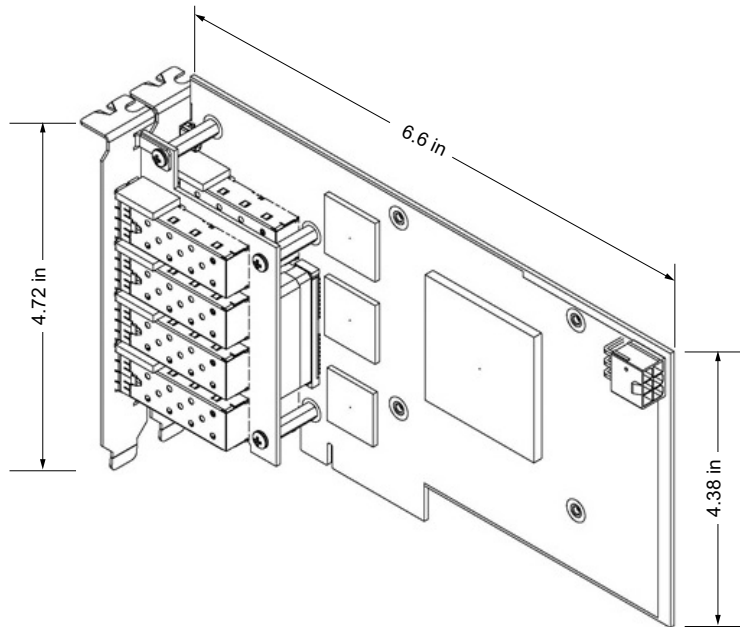


Figure A-1: TILEncore-Gx72 Card Dimensions

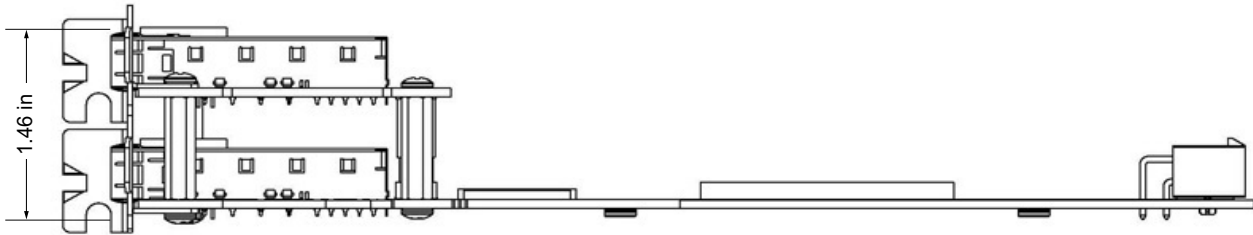


Figure A-2: TILEncore-Gx72 Card Thickness

A.3 Shock and Vibration Specifications

The TILEncore-Gx72 is built to withstand shock and vibration that would be typical in shipping and in non-mobile operating environments. If the board is to be shipped while installed in a host server, it is recommended that the rear bracket be firmly secured into the chassis and that the opposite edge of the card (with the DC power connector) be well supported.

[Table A-3](#) lists the shock and vibration test specifications TILEncore-Gx72 card.

Table A-3. Shock and Vibration Specifications

Specification	Range
Random Vibration	5 - 500 Hz @ 1G rms (9.81 m/s^2 rms), X, Y, and Z axis, 15 min each axis
Shock	Half-Sine @ 80G (785 m/s^2), 2 msec duration, +/- X, Y, Z axis

Detailed test reports can be provided upon request.

A.4 Environmental Specifications

A.4.1 Environment

Table A-4 lists the other environmental specifications of the TILEncore-Gx72 card.

Table A-4. Chassis-Level Environmental Specifications

Specification	Range
Storage temperature	-40° C to +70° C
Operating temperature	0° C to +50° C This represents ambient temperatures surrounding the TILEncore-Gx72 card.
Operating humidity	5% to 95%, non-condensing
Vibration	5 Hz to 20 Hz: 0.01 g ² /Hz sloping up to 0.02 g ² /Hz 20 Hz to 500 Hz: 0.02 g ² /Hz (flat)
Air Flow	200 LFM (passive cooling solution)

Note: The operation of the passive heat sink has been strictly limited to 1G operation with 200 LFM of airflow.

If your airflow values are different, please contact your EZchip support representative for additional information about the impact and functional limitations.

A.4.2 RoHS Statement

The TILEncore-Gx72 is built to comply with the 2002/95/EC European legislation of 2003-02-13, which states that equipment must be free of:

- Lead
- Mercury
- Cadmium
- Hexavalent Chromium
- Polybrominated Biphenyls (PBB)
- Polybrominated Diphenyl Ether (PBDE)

Products free of these materials are known as being RoHS-6 compliant. However, the TILE-Gx process has a small amount of lead contained in the internal die bond for the “flip chip” package. This means that at the board level, this product is specified to be RoHS-5 compliant. However, there are no other devices containing lead on the PCB, and the soldering technology for all components is lead-free.

APPENDIX B: SAFETY AND REGULATORY COMPLIANCE

B.1 Introduction

This chapter explains important safety precautions that you should observe when operating the TILEncore-Gx72™ card.

Note: Follow all warnings and instructions marked on the equipment.

B.2 Safety Precautions

B.2.1 Use Proper Grounding

To avoid electric shock, make sure that the system is connected to an electrical safety (green wire) ground.

B.2.2 Use ESD Protections

The TILEncore-Gx72 contains static-sensitive devices. Static control procedures should be used when installing, removing, and handling all components. Use an antistatic wrist strap at all times. Wrist straps usually come with an alligator clip that you clip to the metal frame of the chassis after the cover is removed.

Never touch the card with any conductive objects.

For additional handling instructions, refer to [Section 4.1.2 “Handling the TILEncore-Gx72 Card”](#) on page 15.

B.2.3 Use Caution When Working Inside a Chassis



Use caution when working inside the computer chassis as touching or bumping the card's rotating blower module can damage it.

B.2.4 Avoid Fumes and Explosives

Do not operate the TILEncore-Gx72 card in the presence of explosives or inflammable gases or fumes.

B.3 Regulatory Compliance

B.3.1 Emissions and Susceptibility Certifications

The TILEncore-Gx72 card is designed for EMI emissions and susceptibility to the following standards:

- FCC part 15, subpart B, Class A (commercial)
- CE EN 55022: 1998 Class A Conducted emissions
Radiated emissions
- CE EN 55024: 1998 Immunity for ITE Amendment A1: 2001
- CE EN 61000-3-2 2000 Harmonic current emissions
- CE EN 61000-3-3 Voltage fluctuations and flicker
- CE IEC 6100-4-2 1995 ESD Air Discharge 8kV Contact discharge 4kV
- CE IEC 6100-4-3 1995 Radiated Immunity (80 - 1000MHz), 3V/m 80% AM by 1kHz
- CE IEC 6100-4-4 1995 EFT/B; Immunity to electrical fast transients; 1kV Power leads
- CE IEC 6100-4-5 1995 Immunity to conducted surges; common mode; 2kV
- CE IEC 6100-4-6 1996 Conducted immunity
- CE IEC 6100-4-11 1994 Voltage dips and short interruptions

B.3.2 FCC Notice

This equipment generates, uses, and can radiate radio frequency energy. If the TILEncore-Gx72 is not installed correctly and used in accordance with the instruction manual, it can cause interference to radio communications.

APPENDIX C: TROUBLESHOOTING

This chapter describes basic methods for troubleshooting your TILEncore-Gx72™ card.

How do I install the PCIe prebooter?

Boot the board using USB as described in [Section 6.2 “Boot Options”](#) on page 36. Connect a serial console device to the UART.

From the console port, type:

```
sbim --info
sbim --install-prebooter /boot/pci-preboot.bin
```

At the prompt, respond with “Yes” and then enter:

```
sbim --info
```

The system responds with something like:

```
SROM booter: Present 8336 bytes
Time Installed: Tue
```

What does `tile-console` do?

The `tile-console` command provides access to the console port on a Tile Processor, which displays Linux boot messages and normally provides an interactive Linux login, via a serial device on the host.

Use the same device name nomenclature as with `tile-monitor`. For example:

```
--dev usb0
—OR—
--dev gxpci0
```

With `tile-btk` operations, I cannot reboot the TILEncore-Gx72 card with `--dev gxpci0` after boot it with `--dev usb0`. Is this expected?

Yes, you can expect this behavior. We recommend power-cycling the host *after* every time you run the `btk` commands.

Even if the `btk` can reboot the TILE-Gx from the SROM and potentially get the PCIe link up, the host does not assign any BAR space to the TILE-Gx, unless the TILEncore-Gx72 supports PCI re-scan, which it does not. Therefore, no further PCI communication can happen.

APPENDIX D: GLOSSARY

Term	Definition
4-WIRE	4-WIRE, or two pair, Synchronous Serial Interface or Serial Peripheral Interface (SPI).
BIB	Board Information Block. The EZchip® run-time software stack uses a set of system-specific information to govern the configuration and initialization of the Tile Processor™ and the devices to which it connects. This data is contained in a structure called the Board Information Block (BIB).
CPLD	Complex Programmable Logic Device
DDR3	Double Data Rate, Type 3. Synchronous dynamic random access memory—a modern kind of dynamic random access memory (DRAM) with a high bandwidth interface.
DIP	Dual In-line Package.
DIMM	Dual In-line Memory Module • LRDIMM – Load reduced • RDIMM – Registered • UDIMM - Unbuffered
EEPROM	Electrically Erasable Programmable Read-Only Memory.
FRU	Field Replaceable Unit.
Gbps	Gigabit per second.
GBps	Gigabyte per second.
GbE	Gigabit Ethernet.
GPIO	General Purpose I/O.
I/O	Input/Output.
JTAG	Joint Test Action Group
LFM	Linear Feet per Minute
LED	Light-Emitting-Diode used for status indicator
LVPECL	Low-Voltage Positive-referenced Emitter-Coupled Logic
MDC/MDIO	Management Clock and Data I/O Interface
MDE™	Multicore Development Environment™. EZchip's multicore programming environment.
PCI	Peripheral Component Interconnect, a bus standard for connecting peripheral devices to a computer motherboard.

Term	Definition
PCIe or PCI-e	PCI Express, a PCI interface format using high-speed serial links for the connection between peripheral devices and the motherboard.
POL	Point Of Load.
POR	Power-On Reset.
ROM	Read-Only Memory.
SERDES	Serializer/Deserializer.
SODIMM	Small Outline DIMM.
SPD ROM	Serial Presence Detect ROM.
SPI	Serial Peripheral Interface.
SPI ROM	Serial Flash with serial peripheral interface.
UART	(Universal Asynchronous Receiver Transmitter). The electronic circuit that makes up the serial port. Also known as “universal serial asynchronous receiver transmitter” (USART), it converts parallel bytes from the CPU into serial bits for transmission, and vice versa. It generates and strips the start and stop bits appended to each character.
USB	Universal Serial Bus
VID	Voltage Identification (bits)
XAUI	10 Gigabit Attachment Unit Interface, a standard for connecting 10 Gigabit Ethernet.
XGbE	10 Gigabit Ethernet

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